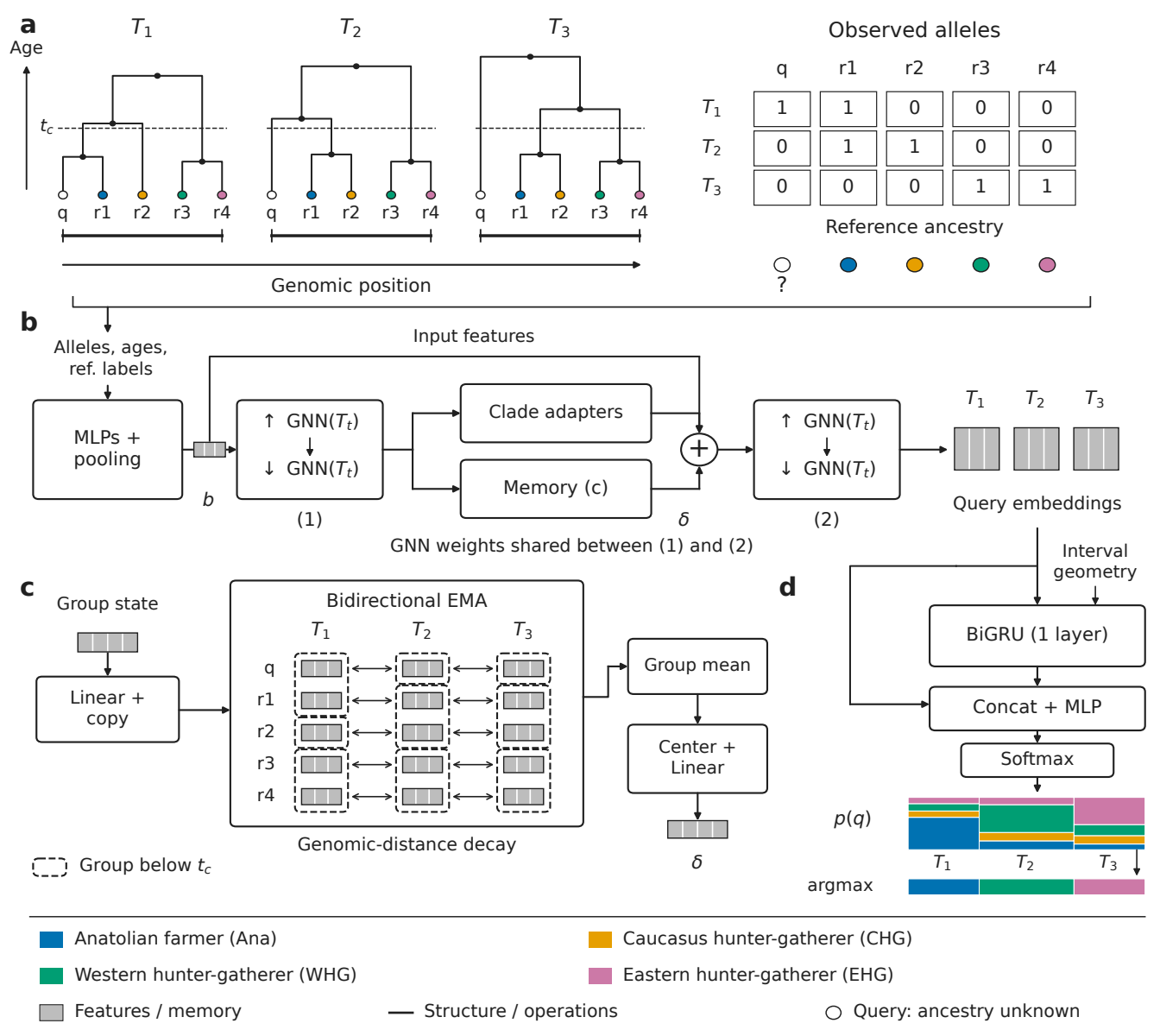
a
Age
T_1
T_2
T_3
t_c
q r1 r2 r3 r4
Genomic position
Observed alleles
q r1 r2 r3 r4
T_1 1 1 0 0 0
T_2 0 1 1 0 0
T_3 0 0 0 1 1
Reference ancestry
?
b
Alleles, ages, ref. labels
MLPs + pooling
Input features
b
↑ GNN(T_t)
↓ GNN(T_t)
(1)
Clade adapters
Memory (c)
δ
↑ GNN(T_t)
↓ GNN(T_t)
(2)
T_1 T_2 T_3
Query embeddings
GNN weights shared between (1) and (2)
c
Group state
Linear + copy
Bidirectional EMA
T_1 T_2 T_3
q r1 r2 r3 r4
Genomic-distance decay
Group mean
Center + Linear
δ
Group below t_c
d
Interval geometry
BiGRU (1 layer)
Concat + MLP
Softmax
$p(q)$
T_1 T_2 T_3
argmax
Anatolian farmer (Ana)
Caucasus hunter-gatherer (CHG)
Western hunter-gatherer (WHG)
Eastern hunter-gatherer (EHG)
Features / memory
Structure / operations
Query: ancestry unknown